EGS 6055 | WEEK 3

Digital Logic Gates

From Binary Thinking to Circuit Design

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EGS 6055 Teaching Demonstration

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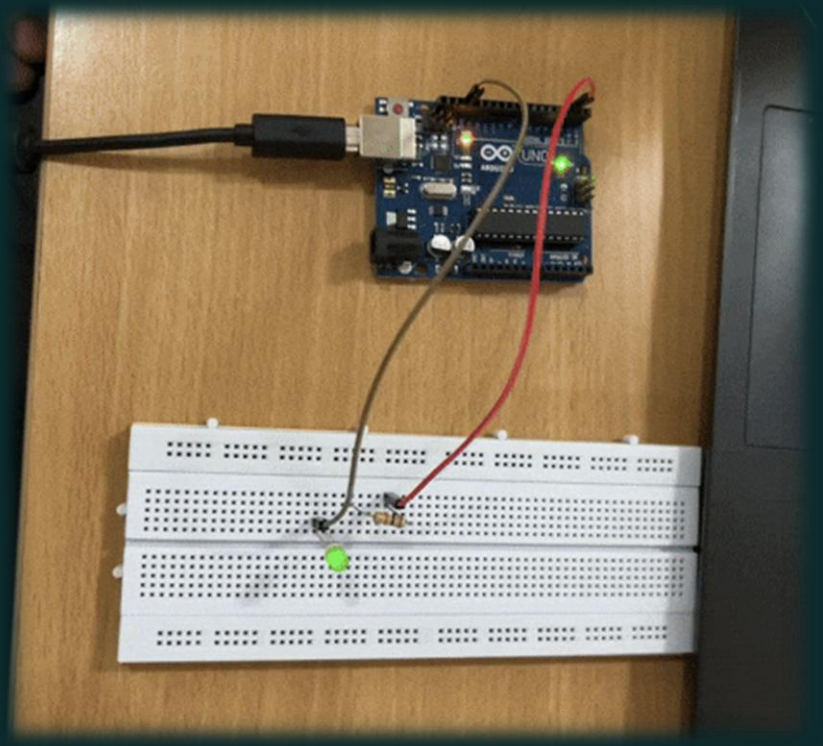


My Teaching Philosophy

01

Belonging First

I believe the first thing a learner needs from me is not a lesson. It is a sign that they are welcome in the classroom. Before I write anything on the board, every student should understand that this shared space is built to support their understanding.

02

Scaffold to the Zone of Proximal Development

Every student I teach is capable of more than they currently know or believe. I look for the edge of what each learner can do with support, then provide guidance that helps them get there. I build in structured reflection so I can tailor my scaffolding individually.

03

Learning as Becoming

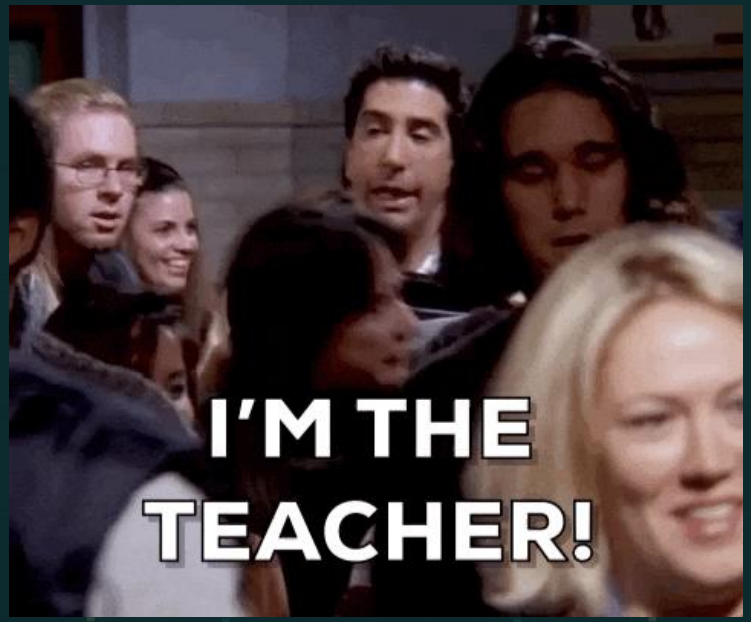
Grounded in Sfard's participation metaphor: My students don't just acquire knowledge, they become participants in engineering practice. Confusion is not failure; it is evidence of learning at the edge of the zone of proximal development.

04

Freirean Problem-Centered Pedagogy

Following Freire, I reject the banking model of education. My teaching centers authentic, real-world problems that require students to act as designers and decision-makers and not passive recipients of correct answers.

Sfard (1998) · Vygotsky ZPD · Lave & Wenger · Brookfield (2017) · Freire (1970)



"The first thing a learner needs from me is not a lesson. It is a sign that they are welcome in the room."

Nicholas Tochukwu Odey

MY THEORY OF LEARNING

Learning is an Act of Becoming

For me, learning is the process by which a person comes to see themselves as a legitimate participant in a community of practice - not merely someone who has acquired information.

Cognitive	What the learner knows and can do
Social	Who they learn with and how
Identity	Whether they believe they belong

Sfard (1998) · Bruner/Vygotsky (1997) · Lave & Wenger - Legitimate Peripheral Participation

Backward Design: Starting With the End in Mind

01 Desired Results

What will students KNOW, DO, and BECOME?

- Identify AND, OR, NOT gates from truth tables & symbols
- Design a combinational circuit for a real-world problem
- Connect Boolean logic to everyday computation
- Students see themselves as legitimate engineering problem-solvers

02 Evidence of Learning

How will I know they got there?

- Live circuit design on **logic.ly**
- Group truth table construction (observed)
- Exit reflection: What did I build? What surprised me?
- Peer explanation of circuit design choices

03 Learning Experiences

What activities lead there?

- Belonging opener: Who designs circuits? Could you?
- Discover-Test-Document cycle with virtual gates and **circuitverse.org**
- Collaborative design challenge using gate knowledge
- Share and critical reflection debrief

Wiggins & McTighe (1998) Understanding by Design · Fink (2003) Significant Learning Experiences

Who Is In The Room?

This course context is for first-year first-semester Electrical and Computer Engineering students, specifically, but open to all tracks of engineering students

The Confident Student

Who they are:

Prior electronics exposure. May conflate logic circuits with electronic circuits.

My teaching move:

Challenge: Design a circuit and prove it with a truth table.

Theory: ZPD: Extend beyond comfort zone

The Cautious Learner

Who they are:

Intellectually capable, finds the subject daunting. Needs belonging before content.

My teaching move:

Hook: You don't need prior experience; you need curiosity.

Theory: Sfard (1998): Invitation into community

The Identity-Challenged

Who they are:

Questions whether they belong in engineering due to representation and social messaging.

My teaching move:

Design prompt: What everyday decision can a logic gate make?

Theory: Henderson et al. (2021): Engineering identity formation

Yadav & Berges (2019) Pedagogical Content Knowledge · Secules et al. (2018) Cultural Construction of Ability

Today's Lesson: 4-Phase Design (45 - Minutes Session, Week 3)

PHASE 1
Belonging Opener
~5 min

- Who designs circuits? Could YOU be one?
- Pair-share: What do you think a logic gate does?
- Establish: Your ideas matter here

PHASE 2
Discover & Document
~10 min

- ICs + breadboards: Test the gate, observe the output
- Groups fill truth table from observations, not lecture
- We collectively name what we discovered

PHASE 3
Design Challenge
~20 min

- Real-world prompt
- Use Logisim (browser) to prototype + breadboard build
- Iterative debugging — confusion is part of engineering

PHASE 4
Reflect & Share
~10 min

- Gallery: Each group explains their design
- Class challenges: Could you do it differently?
- Exit reflection: What did I build? What surprised me?

Smith et al. (2005) Pedagogies of Engagement · Brookfield (2017) Critically Reflective Teacher

One Rule Before We Touch Anything

0

OFF / LOW / FALSE

No voltage signal present

Examples: light off, door closed, no input connected

vs.

1

ON / HIGH / TRUE

Voltage signal is present (5V)

Examples: light on, door open, wire connected to 5V

Every gate decision you will make today is a choice between these two values.

CIRCUITVERSE.ORG

AND Gate – Output = A · B

Guiding Rule

Output is 1 ONLY when BOTH A and B are 1.
If either input is 0, the output is 0.

A	B	A AND B
0	0	0
0	1	0
1	0	0
1	1	1

Real-World Analogy

A security door that opens ONLY when:

- ❑ You have the key card (A = 1)

AND

- ❑ You enter the PIN (B = 1)

Either condition alone is not enough.

Discover Activity – Phase 2

1. Connect your 7408 IC (AND Gate) to the virtual breadboard.
2. Wire both inputs LOW (0V) – observe the LED.
3. Change one input to HIGH (5V) – observe.
4. Change BOTH inputs to HIGH – observe.
5. Fill in your truth table from what you saw.

IC used: 7408 (Quad AND) · Smith et al. (2005) – Active Learning

OR Gate — Output = A + B

The Rule

Output is 1 when AT LEAST ONE input is 1.
Output is 0 ONLY when both inputs are 0.

A	B	A OR B
0	0	0
0	1	1
1	0	1
1	1	1

Real-World Analogy

A home alarm that triggers if:

- ❑ The front door opens (A = 1)

OR

- ❑ The window sensor trips (B = 1)

Only when BOTH are 0 does the alarm stay silent.

Critical Thinking: AND vs OR

AND gate: Output 0 in 3 out of 4 rows
OR gate: Output 1 in 3 out of 4 rows

Ask yourself: *For the home alarm analogy, why would using AND instead of OR be a security flaw?*

IC used: 7432 (Quad OR) · Hammer (1996) - Misconceptions (P-Prims)

NOT Gate + Combining Gates

NOT Gate - Output = $\bar{A}$

Inverts the input. 0 becomes 1; 1 becomes 0.

Input A	NOT A (Output)
0	1
1	0

Analogy: Night light - turns ON when it is NOT bright

What happens when we combine gates?

NAND = NOT after AND
- ❑ True except when BOTH inputs are 1

NOR = NOT after OR
- ❑ True ONLY when both inputs are 0

All complex digital circuits are built from combinations of these basic gates.

The Designer's Question

Before you build any circuit, ask: What decision does my circuit need to make? Then ask: what combination of AND, OR, and NOT gates produces that exact decision?

This is Boolean logic as a design tool.

IC used: 7404 (NOT/Inverter) · Sfard (1998) — Participation Metaphor

Phase 3: The Design Challenge

THE SYSTEM

Design a Hospital Alert System.

The nurse call triggers when: (Patient presses button AND nurse is NOT already in the room) OR (Heart monitor is critical AND patient is NOT in surgery)

Input A

P = Patient

Patient Presses a Button

Input B

N = Nurse

Nurse absent from the room

Input C

H = Heart Monitor

Heart monitor is critical

Input D

S = Surgery

Patient is NOT in surgery

Your Boolean Expression:

$$\text{Hospital Alert System} = (P.N^1)+(H.S^1)$$

Before you build, read each input condition aloud with your group. Does this expression make sense in plain English?

Complete The Truth Table - Hospital Alert System

P Patient	N Nurse	H Heart Monitor	S Surgery	Temp NOT N	Temp NOT S	Temp (P.N[1])	Temp (H.S[1])	Hospital System
0	0	0	0	1	1	0	0	0
0	0	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	1
0	0	1	1	1	0	0	0	0
0	1	0	0	0	1	0	0	0
0	1	0	1	0	0	0	0	0
0	1	1	0	0	1	0	1	1
0	1	1	1	0	0	0	0	0
1	0	0	0	1	1	1	0	1
1	0	0	1	1	0	1	0	1
1	0	1	0	1	1	1	1	1
1	0	1	1	1	0	1	0	1
1	1	0	0	0	1	0	0	0
1	1	0	1	0	0	0	0	0
1	1	1	0	0	1	0	1	1
1	1	1	1	0	0	0	0	0

How to Approach the Design – 4 Steps

STEP 1

Map It

Identify: how many gates do you need?

Calculate the Boolean expression that makes it work

Ask your group:

Can you write this with just one gate type?

STEP 2

Draw It

Fill in the truth table on your handout.

Circle every row where Light = 1.

Ask: Does it match what we said in plain English?

Ask your group:

How many combinations turn the system ON?

STEP 3

Sketch It

Sketch the circuit on paper first.

Label every wire with its value for one test case.

Ask your group:

How did we work as a group to achieve this?

STEP 4

Simulate It

Build in Logisim Evolution (browser).

Toggle all input combinations (A, B, C, D, each 0 or 1).

For each row: does the output match your expression?

Ask your group:

Did the system work as expected

Confusion and debugging are expected and valued. The process is the learning · Brookfield (2017)

Assessment Philosophy: Authentic Evidence of Learning

How I Assess - and Why

Observed Truth Table Build

Formative - Phase 2 | Reveals how students construct knowledge, not just recall it

Breadboard/Logisim Circuit

Performance - Phase 3 | Authentic task aligned to real engineering practice

Exit Reflection Slip

Formative - end of class | What did I build? What surprised me? What am I still unsure of?

Group Design Explanation

Peer - Phase 4 gallery | Articulation deepens understanding; assesses communication + design logic

Equity in Assessment

No timed individual test in Week 3 - students are still building belonging

Multiple assessment modes (writing, building, explaining) honor diverse strengths

I assess the design process, not just the final answer - confusion counts

Reflections are ungraded and confidential

Wiggins (2011) Authentic Assessment · Ko (2021) Equity in Grading · Mitchell & Delaney (2004)

What I Want You To Leave With

Learning is an act of becoming - you don't just learn logic gates, you become someone who reasons with them.

You can build it

By the end of today, every student in this room will have designed a working circuit. Not because it's easy - because we scaffolded it together.

Logic is transferable

Breaking a complex decision into AND/OR/NOT logic is a computational thinking skill that travels beyond ECE or Engineering - into software, into policy, into life.

You belong here

This classroom was designed assuming you would arrive. Your confusion, your creativity, and your questions are the raw material of this lesson.

Thank You !!!!!!!!!!!!!!!!!!!!!!!!!!!!!!